activity 1.py

```python
# Case Study 1: Attendance Recovery Computation
# PDF Topic No. 12

total_classes = int(input("Enter total classes conducted: "))
attended_classes = int(input("Enter classes attended: "))

if total_classes <= 0 or attended_classes < 0 or attended_classes > total_classes:
    print("Invalid input")
else:
    percentage = (attended_classes / total_classes) * 100

    print("Attendance percentage:", round(percentage, 2), "%")

    if percentage >= 75:
        print("You are eligible for the examination.")
        print("No additional classes are required.")
    else:
        additional = 0

        while ((attended_classes + additional) /
               (total_classes + additional)) * 100 < 75:
            additional = additional + 1

```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\JYOTHSNA\OneDrive\Desktop\python assignments> & C:\Users\JYOTHSNA\AppData\Local\Py
p/python assignments/activity 1.py"
Enter total classes conducted: 30
Enter classes attended: 27
Attendance percentage: 90.0 %
You are eligible for the examination.
No additional classes are required.
PS C:\Users\JYOTHSNA\OneDrive\Desktop\python assignments>
```

Search

activity 2.py

```python
# Case Study 2: ATM Transaction Sequence Simulation
# PDF Topic No. 13

WITHDRAWAL_LIMIT = 10000

balance = float(input("Enter opening balance: "))
transaction_count = int(input("Enter number of transactions: "))

if balance < 0 or transaction_count < 0:
    print("Invalid opening balance or transaction count.")
else:
    successful_deposits = 0
    successful_withdrawals = 0
    rejected_transactions = 0
    total_deposited = 0.0
    total_withdrawn = 0.0

    for transaction_number in range(1, transaction_count + 1):
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\JYOTHSNA\OneDrive\Desktop\python assignments> & C:\Users\JYOTHSNA\AppDa
nments/activity 2.py"
Enter opening balance: 20000
Enter number of transactions: 2

Transaction 1
Enter D for deposit or W for withdrawal: W
Enter transaction amount: 10000
Withdrawal successful.
Current balance: Rs. 10000.0

Transaction 2
Enter D for deposit or W for withdrawal: W
Enter transaction amount: 3000
Withdrawal successful.
Current balance: Rs. 7000.0
```

Search

activity 10.py

```python
# Case 10: Food Waste Monitoring System
# Compares meals prepared, served, and left over across several days.

print("FOOD WASTE MONITORING SYSTEM")

while True:
    try:
        days = int(input("Enter number of days to monitor: "))
        if days > 0:
            break
        print("Please enter a positive number of days.")
    except ValueError:
        print("Please enter a whole number.")

while True:
    try:
        limit = float(input("Enter acceptable wastage limit (0-100%): "))
        if 0 <= limit <= 100:
```

Problems Output Debug Console Terminal Ports

```
Day: TUESDAY
Prepared: 4
Served: 3
Leftovers: 1
Wastage percentage: 25.00%
Alert: Wastage exceeds the acceptable limit.
Meal count check: Consistent

========== OVERALL SUMMARY ==========
Total meals prepared: 9
Total meals served: 6
Total leftover meals: 3
Overall wastage percentage: 33.33%
Overall alert: Wastage exceeds the acceptable limit.
Days with meal-count mismatches: 0
Monitoring complete.
```

Search

activity 10.py

```python
# Case 10: Food Waste Monitoring System
# Compares meals prepared, served, and left over across several days.

print("FOOD WASTE MONITORING SYSTEM")

while True:
    try:
        days = int(input("Enter number of days to monitor: "))
        if days > 0:
            break
        print("Please enter a positive number of days.")
    except ValueError:
        print("Please enter a whole number.")

while True:
    try:
        limit = float(input("Enter acceptable wastage limit (0-100%): "))
        if 0 <= limit <= 100:
```

Problems Output Debug Console Terminal Ports

```
Prepared: 5
Served: 3
Leftovers: 2
Wastage percentage: 40.00%
Alert: Wastage exceeds the acceptable limit.
Meal count check: Consistent

Day: TUESDAY
Prepared: 4
Served: 3
Leftovers: 1
Wastage percentage: 25.00%
Alert: Wastage exceeds the acceptable limit.
Meal count check: Consistent

========== OVERALL SUMMARY ==========
```

Search

activity 10.py

```python
# Case 10: Food Waste Monitoring System
# Compares meals prepared, served, and left over across several days.

print("FOOD WASTE MONITORING SYSTEM")

while True:
    try:
        days = int(input("Enter number of days to monitor: "))
        if days > 0:
            break
        print("Please enter a positive number of days.")
    except ValueError:
        print("Please enter a whole number.")

while True:
    try:
        limit = float(input("Enter acceptable wastage limit (0-100%): "))
        if 0 <= limit <= 100:
```

Problems Output Debug Console Terminal Ports

```
Meals prepared: 4
Meals served: 3
Leftover meals: 1

========== DAILY WASTE REPORT ==========

Day: MONDAY
Prepared: 5
Served: 3
Leftovers: 2
Wastage percentage: 40.00%
Alert: Wastage exceeds the acceptable limit.
Meal count check: Consistent

Day: TUESDAY
Prepared: 4
```

Search

activity 10.py

```python
# Case 10: Food Waste Monitoring System
# Compares meals prepared, served, and left over across several days.

print("FOOD WASTE MONITORING SYSTEM")

while True:
    try:
        days = int(input("Enter number of days to monitor: "))
        if days > 0:
            break
        print("Please enter a positive number of days.")
    except ValueError:
        print("Please enter a whole number.")

while True:
    try:
        limit = float(input("Enter acceptable wastage limit (0-100%): "))
        if 0 <= limit <= 100:
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\JYOTHSNA\OneDrive\Desktop\python assignments> & C:\Users\JYOTHSNA\AppData\Local\Py
nments/activity 10.py"
FOOD WASTE MONITORING SYSTEM
Enter number of days to monitor: 2
Enter acceptable wastage limit (0-100%): 2

Day 1
Enter day name/label: MONDAY
Meals prepared: 5
Meals served: 3
Leftover meals: 2

Day 2
Enter day name/label: TUESDAY
Meals prepared: 4
Meals served: 3
```

Search

activity 9.py

```python
# Case Study 9: Tuple-Based Product Catalogue Audit
# Topic 49 - Uses tuples, lists, loops, conditions, and tuple unpacking.

print("=" * 60)
print("            TUPLE-BASED PRODUCT CATALOGUE AUDIT")
print("=" * 60)

while True:
    try:
        product_count = int(input("Enter number of product records: "))
        if product_count > 0:
            break
        print("Enter a positive whole number.")
    except ValueError:
        print("Invalid input. Enter a whole number.")

while True:
    try:
```

Problems Output Debug Console Terminal Ports

```
Total valid inventory value: Rs 15500.00

PRODUCTS REQUIRING REPLENISHMENT (stock below 30)
None

INVALID PRODUCT RECORDS
None

AUDIT SUMMARY
Total records entered: 2
Valid records: 2
Invalid records: 0
Products needing replenishment: 0

Catalogue audit complete.
PS C:\Users\JYOTHSNA\OneDrive\Desktop\python assignments>
```

Search

```python
# Case Study 9: Tuple-Based Product Catalogue Audit
# Topic 49 - Uses tuples, lists, loops, conditions, and tuple unpacking.

print("=" * 60)
print("            TUPLE-BASED PRODUCT CATALOGUE AUDIT")
print("=" * 60)

while True:
    try:
        product_count = int(input("Enter number of product records: "))
        if product_count > 0:
            break
        print("Enter a positive whole number.")
    except ValueError:
        print("Invalid input. Enter a whole number.")

while True:
    try:
```

Problems Output Debug Console Terminal Ports

```
Product record 2
Product ID: 12#4
Product name: GRAINS
Unit price (Rs): 250
Stock quantity: 30

============================================================
VALID PRODUCT CATALOGUE
============================================================
ID          Product                 Price (Rs)    Stock    Value (Rs)
------------------------------------------------------------
1@3         RICE                    200.00        40       8000.00
12#4        GRAINS                  250.00        30       7500.00
------------------------------------------------------------
Total valid inventory value: Rs 15500.00
```

Search

activity 9.py

```python
# Case Study 9: Tuple-Based Product Catalogue Audit
# Topic 49 - Uses tuples, lists, loops, conditions, and tuple unpacking

print("=" * 60)
print("            TUPLE-BASED PRODUCT CATALOGUE AUDIT")
print("=" * 60)

while True:
    try:
        product_count = int(input("Enter number of product records: "))
        if product_count > 0:
            break
        print("Enter a positive whole number.")
    except ValueError:
        print("Invalid input. Enter a whole number.")

while True:
    try:
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\JYOTHSNA\OneDrive\Desktop\python assignments> & C:\Users\JYOTHSNA\AppData\
nments/activity 9.py"
============================================================
            TUPLE-BASED PRODUCT CATALOGUE AUDIT
============================================================
Enter number of product records: 2
Enter minimum stock level (reorder threshold): 30

Product record 1
Product ID: 1@3
Product name: RICE
Unit price (Rs): 200
Stock quantity: 40

Product record 2
Product ID: 12#4
```

Search

activity 8.py

```python
# Case Study 8: Student Marks Matrix Analysis
# Topic 42 - Uses nested lists, loops, conditions, and arithmetic.

print("=" * 58)
print("           STUDENT MARKS MATRIX ANALYSIS")
print("=" * 58)

while True:
    try:
        student_count = int(input("Enter number of students: "))
        if student_count > 0:
            break
        print("Please enter a number greater than zero.")
    except ValueError:
        print("Invalid input. Enter a whole number.")

while True:
    try:
```

Problems Output Debug Console Terminal Ports

```
SUBJECT-WISE RESULTS
==========================================================
Subject                  Total     Average
----------------------------------------
PYTHON                 229.00       76.33
ENGLISH                262.00       87.33

TOP PERFORMER(S)
MEGHA
Highest student average: 94.00%

SUBJECTS WITH AVERAGE BELOW 65%
None

Analysis complete.
```

Search

activity 8.py

```python
# Case Study 8: Student Marks Matrix Analysis
# Topic 42 - Uses nested lists, loops, conditions, and arithmetic.

print("=" * 58)
print("          STUDENT MARKS MATRIX ANALYSIS")
print("=" * 58)

while True:
    try:
        student_count = int(input("Enter number of students: "))
        if student_count > 0:
            break
        print("Please enter a number greater than zero.")
    except ValueError:
        print("Invalid input. Enter a whole number.")

while True:
    try:
```

Problems Output Debug Console Terminal Ports

```
==========================================================
STUDENT-WISE RESULTS
==========================================================
Student                 Total       Average
----------------------------------------
RAVI                   118.00         59.00
MEGHA                  188.00         94.00
SANA                   185.00         92.50

SUBJECT-WISE RESULTS
==========================================================
Subject                 Total       Average
----------------------------------------
PYTHON                 229.00         76.33
ENGLISH                262.00         87.33
```

Search

activity 8.py

```python
# Case Study 8: Student Marks Matrix Analysis
# Topic 42 - Uses nested lists, loops, conditions, and arithmetic.

print("=" * 58)
print("          STUDENT MARKS MATRIX ANALYSIS")
print("=" * 58)

while True:
    try:
        student_count = int(input("Enter number of students: "))
        if student_count > 0:
            break
        print("Please enter a number greater than zero.")
    except ValueError:
        print("Invalid input. Enter a whole number.")

while True:
    try:
```

Problems Output Debug Console Terminal Ports

```
Enter name of student 2: MEGHA
Enter PYTHON marks for MEGHA (0-100): 90
Enter ENGLISH marks for MEGHA (0-100): 98

Enter name of student 3: SANA
Enter PYTHON marks for SANA (0-100): 89
Enter ENGLISH marks for SANA (0-100): 96

==========================================================
STUDENT-WISE RESULTS
==========================================================
Student                  Total      Average
----------------------------------------
RAVI                     118.00       59.00
MEGHA                    188.00       94.00
```

Search

activity 1.py | activity 4.py | activity 8.py | activity 7.py | activity 6.py | activity 5.py

python assignments > activity 8.py > ...

```python
# Case Study 8: Student Marks Matrix Analysis
# Topic 42 - Uses nested lists, loops, conditions, and arithmetic.

print("=" * 58)
print("          STUDENT MARKS MATRIX ANALYSIS")
print("=" * 58)

while True:
    try:
        student_count = int(input("Enter number of students: "))
        if student_count > 0:
            break
        print("Please enter a number greater than zero.")
    except ValueError:
        print("Invalid input. Enter a whole number.")

while True:
    try:
```

Problems | Output | Debug Console | Terminal | Ports

```
PS C:\Users\JYOTHSNA\OneDrive\Desktop\python assignments> & C:\Users\JYOTHSNA\AppData\Local\Python\pyth
nments/activity 8.py"
==========================================================
          STUDENT MARKS MATRIX ANALYSIS
==========================================================
Enter number of students: 3
Enter number of subjects: 2
Enter name of subject 1: PYTHON
Enter name of subject 2: ENGLISH

Enter name of student 1: RAVI
Enter PYTHON marks for RAVI (0-100): 50
Enter ENGLISH marks for RAVI (0-100): 68

Enter name of student 2: MEGHA
Enter PYTHON marks for MEGHA (0-100): 90
```

Search

```python
"""
Case Study 7: String-Based Examination Answer Evaluation
Compare student answer strings with an answer key, score responses,
apply negative marking, and display a result summary.
"""

print("STRING-BASED EXAMINATION ANSWER EVALUATION")
print("-----------------------------------------")

while True:
    try:
        question_count = int(input("Enter number of questions: "))
        if question_count > 0:
            break
        print("Enter a number greater than zero.")
    except ValueError:
        print("Invalid input. Enter a whole number.")

answer_key = []
```

Problems Output Debug Console **Terminal** Ports

```
Enter student name: MEGHA
Answer Q1 (A/B/C/D or Enter to skip): A
Answer Q2 (A/B/C/D or Enter to skip): C
Answer Q3 (A/B/C/D or Enter to skip): B
Answer Q4 (A/B/C/D or Enter to skip): D
Answer Q5 (A/B/C/D or Enter to skip): B

Result for MEGHA
Correct answers: 3
Incorrect answers: 2
Unanswered questions: 0
Final score: 2.5 out of 5
Score percentage: 50.0 %

Evaluation completed.
PS C:\Users\JYOTHSNA\OneDrive\Desktop\python assignments>
```

Search

```python
"""
Case Study 7: String-Based Examination Answer Evaluation
Compare student answer strings with an answer key, score responses,
apply negative marking, and display a result summary.
"""

print("STRING-BASED EXAMINATION ANSWER EVALUATION")
print("-----------------------------------------")

while True:
    try:
        question_count = int(input("Enter number of questions: "))
        if question_count > 0:
            break
        print("Enter a number greater than zero.")
    except ValueError:
        print("Invalid input. Enter a whole number.")

answer_key = []
```

Problems Output Debug Console **Terminal** Ports

```
Answer Q4 (A/B/C/D or Enter to skip): C
Answer Q5 (A/B/C/D or Enter to skip): A

Result for RAVI
Correct answers: 3
Incorrect answers: 2
Unanswered questions: 0
Final score: 2.5 out of 5
Score percentage: 50.0 %

Student 2
Enter student name: MEGHA
Answer Q1 (A/B/C/D or Enter to skip): A
Answer Q2 (A/B/C/D or Enter to skip): C
Answer Q3 (A/B/C/D or Enter to skip): B
Answer Q4 (A/B/C/D or Enter to skip): D
```

Search

activity 7.py

python assignments > activity 7.py > ...

```python
"""
Case Study 7: String-Based Examination Answer Evaluation
Compare student answer strings with an answer key, score responses,
apply negative marking, and display a result summary.
"""

print("STRING-BASED EXAMINATION ANSWER EVALUATION")
print("----------------------------------------")

while True:
    try:
        question_count = int(input("Enter number of questions: "))
        if question_count > 0:
            break
        print("Enter a number greater than zero.")
    except ValueError:
        print("Invalid input. Enter a whole number.")

answer_key = []
```

Problems Output Debug Console Terminal Ports

```
STRING-BASED EXAMINATION ANSWER EVALUATION
----------------------------------------
Enter number of questions: 5
Enter correct answer for question 1 (A/B/C/D): A
Enter correct answer for question 2 (A/B/C/D): C
Enter correct answer for question 3 (A/B/C/D): D
Enter correct answer for question 4 (A/B/C/D): C
Enter correct answer for question 5 (A/B/C/D): B
Enter number of students: 2

Student 1
Enter student name: RAVI
Answer Q1 (A/B/C/D or Enter to skip): A
Answer Q2 (A/B/C/D or Enter to skip): B
Answer Q3 (A/B/C/D or Enter to skip): D
Answer Q4 (A/B/C/D or Enter to skip): C
```

Search

activity 6.py

```python
"""
Case Study 6: Password Policy Validation Engine
Checks password length, character categories,
repetition, and prohibited substrings.
"""

print("PASSWORD POLICY VALIDATION ENGINE")
print("---------------------------------")

MIN_LENGTH = 8
PROHIBITED_SUBSTRINGS = ["password", "admin", "1234"]

while True:
    try:
        count = int(input("How many passwords do you want to check? "))
        if count > 0:
            break
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\JYOTHSNA\OneDrive\Desktop\python assignments> & C:\Users\JYOTHSNA\AppData
nments/activity 6.py"
PASSWORD POLICY VALIDATION ENGINE
---------------------------------
How many passwords do you want to check? 2

Enter password 1: Jsd1@45_67*9
Result:
VALID - Password satisfies all policy rules.

Enter password 2: jAr56_%e6
Result:
VALID - Password satisfies all policy rules.

Password checking completed.
PS C:\Users\JYOTHSNA\OneDrive\Desktop\python assignments>
```

Search

```python
"""
Case Study 5: Advanced Numeric Pattern Generator
Generate a hollow diamond using nested loops and validate the request
"""

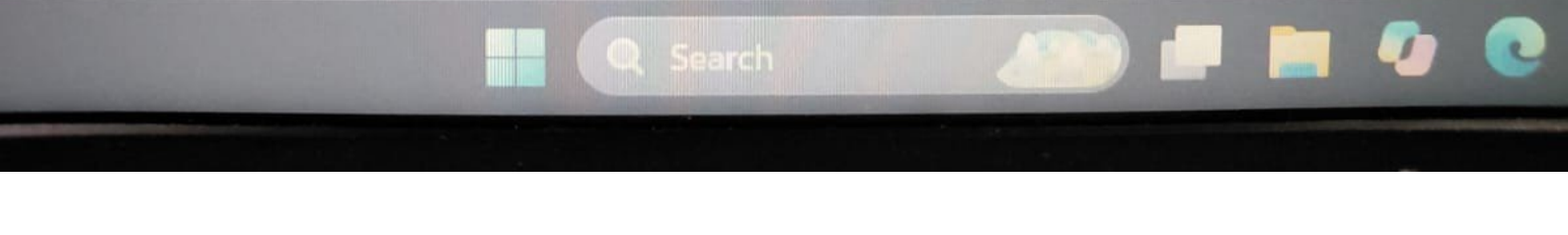
print("ADVANCED NUMERIC PATTERN GENERATOR")
print("----------------------------------")

while True:
    try:
        size = int(input("Enter diamond size (1-9): "))
        if 1 <= size <= 9:
            break
        print("Invalid size. Enter a number from 1 to 9.")
    except ValueError:
        print("Invalid input. Please enter a whole number.")

# Upper half, including the middle row
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\JYOTHSNA\OneDrive\Desktop\python assignments> & C:\Users\JYOTHSNA\Appl
nments/activity 5.py"
ADVANCED NUMERIC PATTERN GENERATOR
----------------------------------
Enter diamond size (1-9): 3
  1
 2 2
3   3
 2 2
  1
Pattern generated successfully.
PS C:\Users\JYOTHSNA\OneDrive\Desktop\python assignments>
```



activity 4.py

```python
"""
Case Study 4: Multi-Product Shopping Budget Constraint
Evaluates a shopping basket, applies discounts, and reduces quantities
until the total fits within the budget.
"""

print("MULTI-PRODUCT SHOPPING BUDGET PLANNER")

budget = float(input("Enter your shopping budget (Rs): "))

while budget < 0:
    print("Budget cannot be negative.")
    budget = float(input("Enter your shopping budget (Rs): "))

count = int(input("Enter number of different products: "))

while count <= 0:
```

Problems Output Debug Console Terminal Ports

```
Discount: Rs 0.0
Total after discount: Rs 400.0
Budget: Rs 400.0

--- RECOMMENDED BASKET ---
Wheat floor packet : 1 x Rs 200.0 = Rs 200.0
oil packet : 1 x Rs 200.0 = Rs 200.0
Revised subtotal: Rs 400.0
Discount rate: 0.0 %
Discount: Rs 0.0
Final total: Rs 400.0
Money remaining: Rs 0.0

Original basket is already within budget.
Status: Basket fits within budget.
PS C:\Users\JYOTHSNA\OneDrive\Desktop\python assignments>
```

Search

activity 4.py

```python
"""
Case Study 4: Multi-Product Shopping Budget Constraint
Evaluates a shopping basket, applies discounts, and reduces quantities
until the total fits within the budget.
"""

print("MULTI-PRODUCT SHOPPING BUDGET PLANNER")

budget = float(input("Enter your shopping budget (Rs): "))

while budget < 0:
    print("Budget cannot be negative.")
    budget = float(input("Enter your shopping budget (Rs): "))

count = int(input("Enter number of different products: "))

while count <= 0:
```

Problems Output Debug Console Terminal Ports

```
Quantity in just numbers: 1

--- ORIGINAL BASKET ---
Wheat floor packet : 1 x Rs 200.0 = Rs 200.0
oil packet : 1 x Rs 200.0 = Rs 200.0
Subtotal: Rs 400.0
Discount rate: 0.0 %
Discount: Rs 0.0
Total after discount: Rs 400.0
Budget: Rs 400.0

--- RECOMMENDED BASKET ---
Wheat floor packet : 1 x Rs 200.0 = Rs 200.0
oil packet : 1 x Rs 200.0 = Rs 200.0
Revised subtotal: Rs 400.0
Discount rate: 0.0 %
```

Search

```python
"""
Case Study 4: Multi-Product Shopping Budget Constraint
Evaluates a shopping basket, applies discounts, and reduces quantities
until the total fits within the budget.
"""

print("MULTI-PRODUCT SHOPPING BUDGET PLANNER")

budget = float(input("Enter your shopping budget (Rs): "))

while budget < 0:
    print("Budget cannot be negative.")
    budget = float(input("Enter your shopping budget (Rs): "))

count = int(input("Enter number of different products: "))

while count <= 0:
```

Problems Output Debug Console **Terminal** Ports

```
PS C:\Users\JYOTHSNA\OneDrive\Desktop\python assignments> & C:\Users\JYOTHSNA\AppData\Local\Pyth
nments/activity 4.py"
MULTI-PRODUCT SHOPPING BUDGET PLANNER
Enter your shopping budget (Rs): 400
Enter number of different products: 2

Product 1
Product name: Wheat floor packet
Unit price (Rs): 200
Quantity in just numbers: 1

Product 2
Product name: oil packet
Unit price (Rs): 200
Quantity in just numbers: 1
```

Search

activity 3.py

```python
# Case Study 3: Student Performance Trend Analysis
# PDF Topic No. 17

subject_count = int(input("Enter number of subjects: "))
exam_count = 3

if subject_count <= 0:
    print("Invalid input: number of subjects must be greater t
else:
    subjects = []
    marks_list = []

    for i in range(subject_count):
        subject = input("\nEnter subject name: ").strip()

        while subject == "":
            print("Subject name cannot be empty.")
            subject = input("Enter subject name: ").strip()
```

Problems Output Debug Console Terminal Ports

```
Subject: Mathematics
Average marks: 90.0
Change from Exam 1 to Exam 3: -1.0
Trend: Declined
Performance category: Excellent

Subject: python
Average marks: 91.67
Change from Exam 1 to Exam 3: 7.0
Trend: Improved
Performance category: Excellent

Overall average: 92.89 %
Overall classification: Excellent
No subjects are below the 40% average threshold.
PS C:\Users\JYOTHSNA\OneDrive\Desktop\python assignments>
```

activity 3.py

python assignments > activity 3.py > ...

```python
# Case Study 3: Student Performance Trend Analysis
# PDF Topic No. 17

subject_count = int(input("Enter number of subjects: "))
exam_count = 3

if subject_count <= 0:
    print("Invalid input: number of subjects must be greater t
else:
    subjects = []
    marks_list = []

    for i in range(subject_count):
        subject = input("\nEnter subject name: ").strip()

        while subject == "":
            print("Subject name cannot be empty.")
            subject = input("Enter subject name: ").strip()
```

Problems Output Debug Console Terminal Ports

```
--- Student Performance Report ---

Subject: English
Average marks: 97.0
Change from Exam 1 to Exam 3: -2.0
Trend: Declined
Performance category: Excellent

Subject: Mathematics
Average marks: 90.0
Change from Exam 1 to Exam 3: -1.0
Trend: Declined
Performance category: Excellent

Subject: python
Average marks: 91.67
```

activity 3.py

```python
# Case Study 3: Student Performance Trend Analysis
# PDF Topic No. 17

subject_count = int(input("Enter number of subjects: "))
exam_count = 3

if subject_count <= 0:
    print("Invalid input: number of subjects must be greater
else:
    subjects = []
    marks_list = []

    for i in range(subject_count):
        subject = input("\nEnter subject name: ").strip()

        while subject == "":
            print("Subject name cannot be empty.")
            subject = input("Enter subject name: ").strip()
```

Problems Output Debug Console Terminal Ports

```
Enter subject name: Mathematics
Enter marks for Mathematics in Exam 1 (0-100): 90
Enter marks for Mathematics in Exam 2 (0-100): 91
Enter marks for Mathematics in Exam 3 (0-100): 89

Enter subject name: python
Enter marks for python in Exam 1 (0-100): 89
Enter marks for python in Exam 2 (0-100): 90
Enter marks for python in Exam 3 (0-100): 96

--- Student Performance Report ---

Subject: English
Average marks: 97.0
Change from Exam 1 to Exam 3: -2.0
```

activity 3.py

```python
# Case Study 3: Student Performance Trend Analysis
# PDF Topic No. 17

subject_count = int(input("Enter number of subjects: "))
exam_count = 3

if subject_count <= 0:
    print("Invalid input: number of subjects must be greater t
else:
    subjects = []
    marks_list = []

    for i in range(subject_count):
        subject = input("\nEnter subject name: ").strip()

        while subject == "":
            print("Subject name cannot be empty.")
            subject = input("Enter subject name: ").strip()
```

Problems Output Debug Console Terminal Ports

```
nments/activity 3.py"
Enter number of subjects: 3

Enter subject name: English
Enter marks for English in Exam 1 (0-100): 98
Enter marks for English in Exam 2 (0-100): 97
Enter marks for English in Exam 3 (0-100): 96

Enter subject name: Mathematics
Enter marks for Mathematics in Exam 1 (0-100): 90
Enter marks for Mathematics in Exam 2 (0-100): 91
Enter marks for Mathematics in Exam 3 (0-100): 89

Enter subject name: python
Enter marks for python in Exam 1 (0-100): 89
Enter marks for python in Exam 2 (0-100): 90
```

activity 3.py

python assignments > activity 3.py > ...

```python
# Case Study 3: Student Performance Trend Analysis
# PDF Topic No. 17

subject_count = int(input("Enter number of subjects: "))
exam_count = 3

if subject_count <= 0:
    print("Invalid input: number of subjects must be greater than zero.")
else:
    subjects = []
    marks_list = []

    for i in range(subject_count):
        subject = input("\nEnter subject name: ").strip()

        while subject == "":
            print("Subject name cannot be empty.")
            subject = input("Enter subject name: ").strip()
```

Problems Output Debug Console Terminal Ports

```
Subject: Mathematics
Average marks: 90.0
Change from Exam 1 to Exam 3: -1.0
Trend: Declined
Performance category: Excellent

Subject: python
Average marks: 91.67
Change from Exam 1 to Exam 3: 7.0
Trend: Improved
Performance category: Excellent

Overall average: 92.89 %
Overall classification: Excellent
No subjects are below the 40% average threshold.
PS C:\Users\JYOTHSNA\OneDrive\Desktop\python assignments>
```

Search

```python
WITHDRAWAL_LIMIT = 10000

balance = float(input("Enter opening balance: "))
transaction_count = int(input("Enter number of transactions: "))

if balance < 0 or transaction_count < 0:
    print("Invalid opening balance or transaction count.")
else:
    successful_deposits = 0
    successful_withdrawals = 0
    rejected_transactions = 0
    total_deposited = 0.0
    total_withdrawn = 0.0

    for transaction_number in range(1, transaction_count + 1):
```

Problems Output Debug Console **Terminal** Ports

```
Enter D for deposit or W for withdrawal: W
Enter transaction amount: 3000
Withdrawal successful.
Current balance: Rs. 7000.0

--- Transaction Summary ---
Successful deposits: 0
Successful withdrawals: 2
Rejected transactions: 0
Total amount deposited: Rs. 0.0
Total amount withdrawn: Rs. 13000.0
Final balance: Rs. 7000.0
PS C:\Users\JYOTHSNA\OneDrive\Desktop\python assignments>
```

Search

activity 1.py | activity 2.py | activity 4.py | activity 6.py | activity 7.py

```python
WITHDRAWAL_LIMIT = 10000

balance = float(input("Enter opening balance: "))
transaction_count = int(input("Enter number of transactions: "))

if balance < 0 or transaction_count < 0:
    print("Invalid opening balance or transaction count.")
else:
    successful_deposits = 0
    successful_withdrawals = 0
    rejected_transactions = 0
    total_deposited = 0.0
    total_withdrawn = 0.0

    for transaction_number in range(1, transaction_count + 1):
```

Problems | Output | Debug Console | Terminal | Ports

```
Enter D for deposit or W for withdrawal: W
Enter transaction amount: 10000
Withdrawal successful.
Current balance: Rs. 10000.0

Transaction 2
Enter D for deposit or W for withdrawal: W
Enter transaction amount: 3000
Withdrawal successful.
Current balance: Rs. 7000.0

--- Transaction Summary ---
Successful deposits: 0
Successful withdrawals: 2
Rejected transactions: 0
Total amount deposited: Rs. 0.0
```

Search